

SpaceX Falcon 9 Project

IBM Data Science Professional
Certificate

<https://github.com/Leviathan-404/SpaceX-Falcon-9-Project>

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OUTLINE



- Executive Summary
- Introduction
- Methodology
- Results
 - Visualization – Charts
 - Dashboard
- Discussion
 - Findings & Implications
- Conclusion



EXECUTIVE SUMMARY



- Analyzed rocket launch data using python
- Used a downloaded dataset because the API was unavailable
- Included a web-scraping notebook to show the alternative data-collection method
- Cleaned and explored the data using pandas and numpy
- Found that landing success varied by booster version and launch site
- Future work could include newer data and predictive modeling

INTRODUCTION



- This capstone project applies the end-to-end data science workflow to a real-world aerospace problem: predicting whether the SpaceX Falcon 9 first stage will land successfully.
- Because first-stage reuse can significantly reduce launch costs, accurate landing predictions can support pricing and competitive bidding decisions.
- The project uses data collection, wrangling, exploratory analysis, visualization, and machine learning to identify factors linked to successful landings.
- The final model evaluation compares predictive performance and highlights insights that can help inform launch cost estimates.



CAPSTONE METHODOLOGY



- Data collection: Scraped Falcon 9 and Falcon Heavy historical launch records from Wikipedia.
-
- EDA + labeling: Explored landing outcomes and converted results into supervised learning labels: 1 = successful landing, 0 = unsuccessful landing.
-
- SQL analysis: Used SQL queries to complete required analysis tasks and summarize launch patterns.
-
- Feature engineering: Cleaned variables and prepared features for model training.
-
- Interactive analytics: Built visual analysis with Folium maps and an interactive Dash app.
-
- Machine learning: Created a pipeline to predict first-stage landing success using the prepared launch data.



DATA COLLECTION & WRANGLING

Source: Wikipedia page, List of Falcon 9 and Falcon Heavy launches.

Collection: Used requests, BeautifulSoup, and pandas to scrape and structure historical launch records after the SpaceX API was unavailable.

Fields captured: Included launch date, booster/version details, payload/orbit information, launch site, landing outcome, reuse indicators, landing pad, and location-related attributes.

Missing-value review: Identified missing attributes and calculated the percentage of missing values in each column.

Wrangling: Standardized landing outcomes, selected modeling features, and converted results into a binary target label: 1 = successful landing, 0 = unsuccessful landing.

DATA PREPARATION PIPELINE

FINAL OUTPUT

Cleaned Falcon 9 launch dataset prepared for:

- Exploratory data analysis
- SQL analysis
- Visual analytics
- Machine learning



SQL ANALYSIS OF LAUNCH RECORDS

SETUP

Tool environment

Jupyter Notebook SQL using %sql magic commands connected to a SQLite3 database.

Dataset

Cleaned Falcon 9 launch records prepared from the Wikipedia web-scraping and wrangling stage.

Purpose

Use SQL to summarize launch data, validate patterns, and prepare insights before visualization and machine learning.

QUERY FOCUS

Main analysis tasks

- Queried launch sites and mission records
- Compared payload mass across customers and boosters
- Reviewed booster versions and landing outcomes
- Counted successful vs. unsuccessful landings
- Used filters, grouping, averages, and date-based queries

Result

SQL helped uncover structured patterns that supported later EDA, Folium/Dash visual analytics, and model feature selection.





SQL EDA: Launch Sites, Mission Outcomes & Payloads

Exploring mission data to uncover key patterns and performance insights



4

Unique Launch Sites



98

Successful Missions



1

Failed Mission
(in flight)



2

Other / Unclear Outcomes
(e.g., payload status unclear)

Mission Outcomes



- Success
98 (97%)
- Failure (in flight)
1 (1%)
- Success
(payload status unclear)
2 (2%)

Payload Analysis



2,928.4 kg

Average Payload Mass

Selected booster:
F9 v1.1



Launch Sites

Unique sites identified (4)

- 1 CCAFS LC-40
- 2 VAFB SLC-4E
- 3 KSC LC-39A
- 4 CCAFS SLC-40



22 Dec 2015



First Successful Ground-Pad Landing



Key Insight

The EDA shows a strong concentration of successful missions, while the first successful ground-pad landing in December 2015 marks an important milestone in booster recovery.



Task 1: Launch Sites

Visualizing the relationship between Flight Number and Launch Site



Key Takeaways

1

Vandenberg SLC-4

The first launch site is Vandenberg Space Launch Complex 4, located in Los Angeles, California.

2

Cape Canaveral SFS (SLC-40)

Two launches occurred at the Cape Canaveral Space Force Station (SLC-40) in Florida.

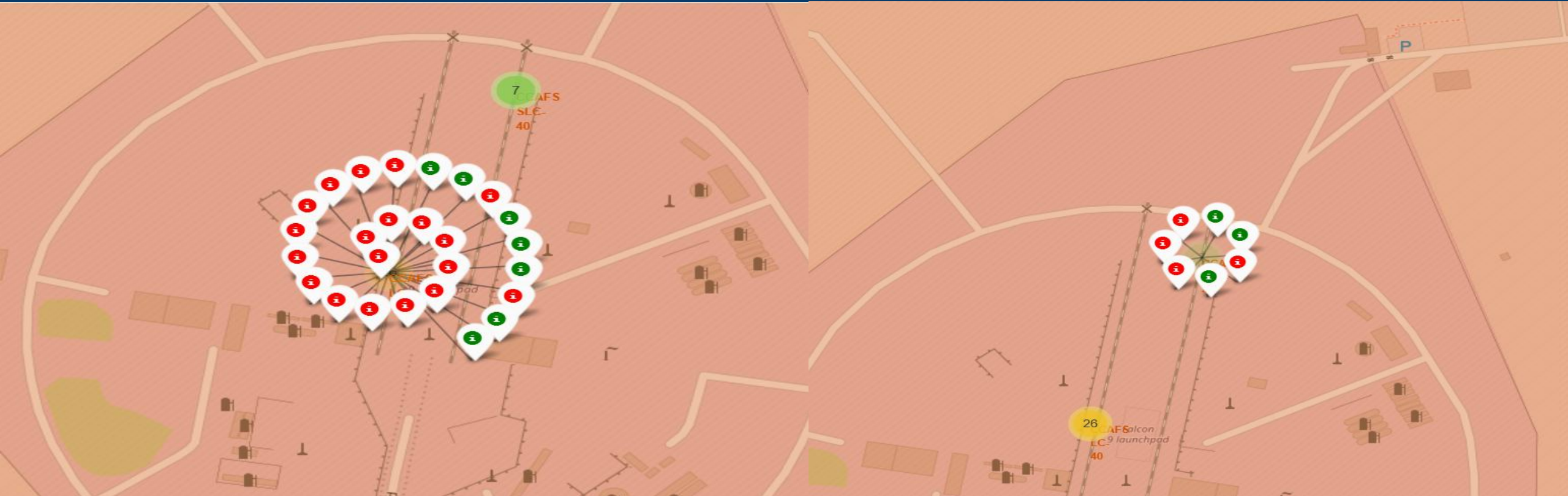
3

Kennedy Parkway North

The third launch site is also near the Cape Canaveral Space Force Station, near Kennedy Parkway North in Florida.



Task 2: Marked success/failed launches for CCFAS SLC-40



Key Takeaways

The two Cape Canaveral-area launch sites show different success rates, with the second site recording a higher proportion of successful launches in this dataset.

1

Key Finding 1

CCAFS SLC-40 (left): Of the **26** recorded launches, **7** were **successful**, giving a success rate of approximately **26.9%**.

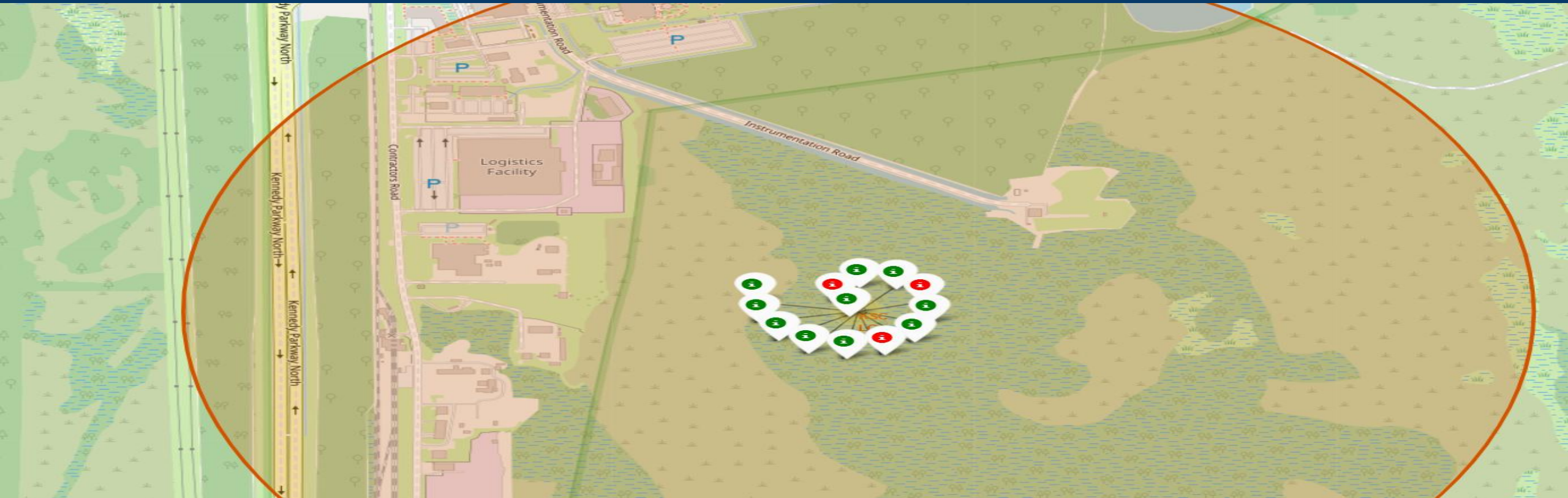
2

Key Finding 2

Second launch site (right): Of **7** recorded launches, **3** were **successful**, giving a success rate of approximately **42.9%**



Task 2: Marked success/failed launches for KSC LC 39A

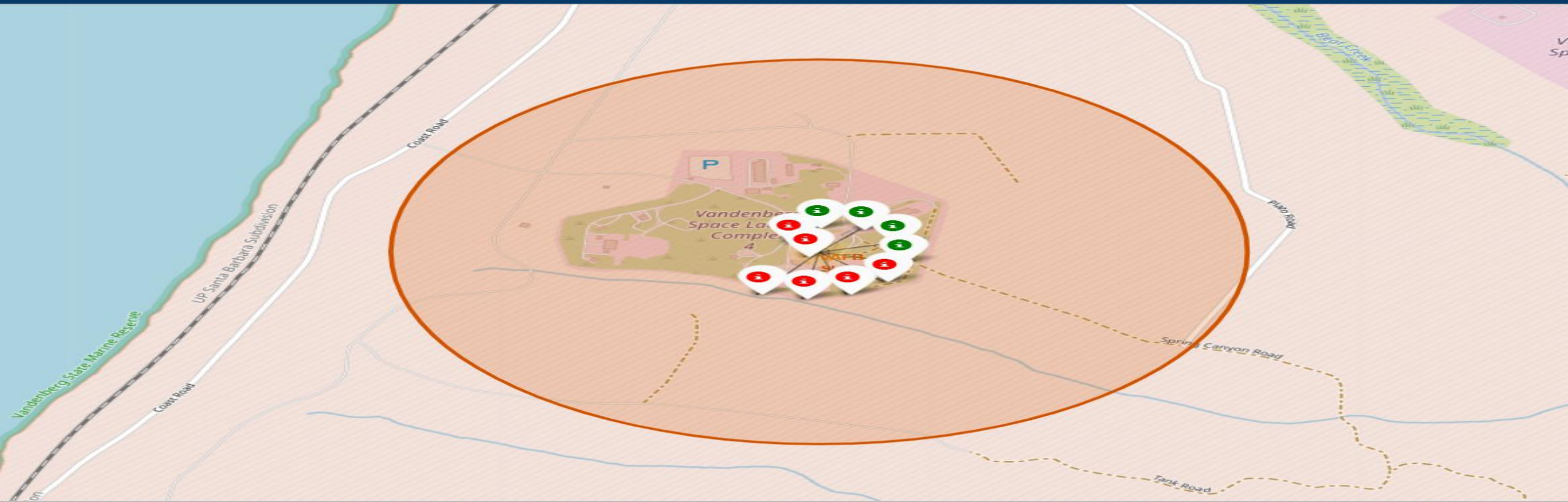


Key Takeaways

Of the **13 launches recorded at KSC LC-39A**, only **3 were unsuccessful**, meaning **10 launches were successful**. This corresponds to a **76.9% success rate**, indicating a relatively high proportion of successful missions at this launch site within the dataset.



Task 2: Marked success/failed launches for KSC LC 39A



Key Takeaways

Of the **10 launches recorded at VAFB SLC-4E in Los Angeles**, **4 were successful**, resulting in a **40% success rate**. The data therefore shows that **less than half of the launches at this site were successful** within the dataset.

Launch Success Analysis

Summary of launch performance by site, payload range, and Falcon 9 booster version

1

Largest share of successful launches

KSC LC-39A — 41.7%

KSC LC-39A accounted for the largest proportion of successful launches, followed by CCAFS LC-40 at 29.2%.

2

Highest launch success rate

KSC LC-39A — 76.9%

10 of 13 launches at KSC LC-39A were successful, producing the highest site-level success rate in the dataset.

3

Payload range with highest success

0–6,000 kg

The displayed payload analysis indicates stronger successful-launch representation within the 0–6,000 kg range.

4

Payload range with lowest success

5,000–9,000 kg

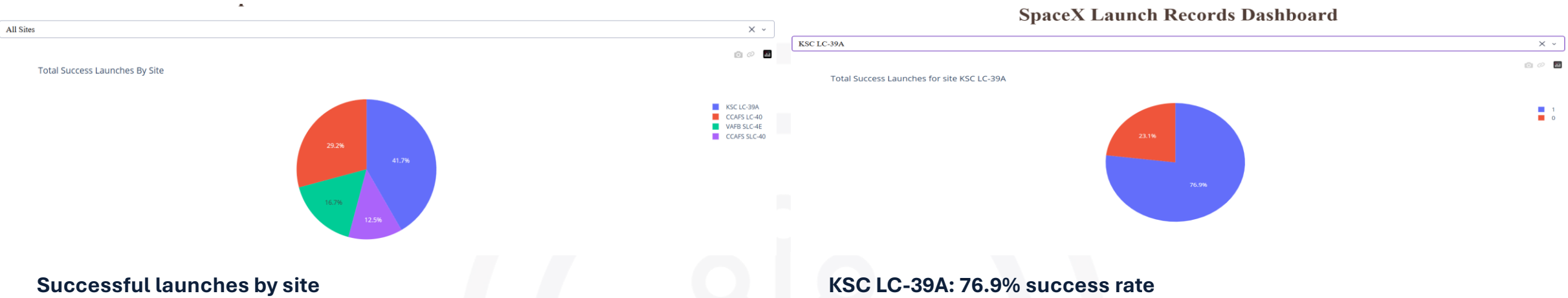
The displayed comparison shows comparatively weaker launch outcomes within the 5,000–9,000 kg range.

5 | Highest-performing booster version: FT — the strongest launch success performance among the displayed booster categories.



Supporting Charts: Launch Site Performance

Dashboard visualizations supporting the site-level findings

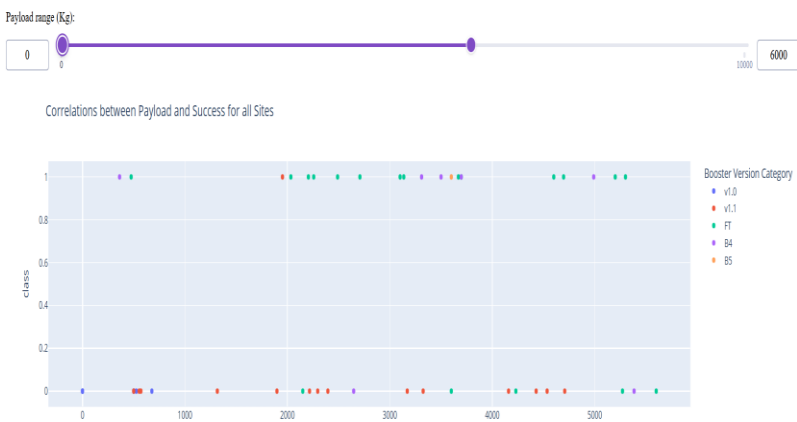


Interpretation

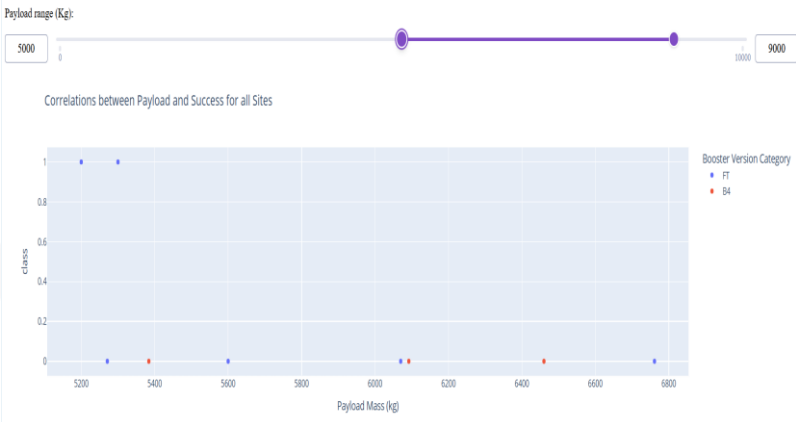
KSC LC-39A leads both in the share of successful launches and in overall launch success rate. The site contributed 41.7% of successful launches and achieved a 76.9% success rate, based on 10 successful launches out of 13 recorded missions.

Supporting Charts: Payload and Booster Performance

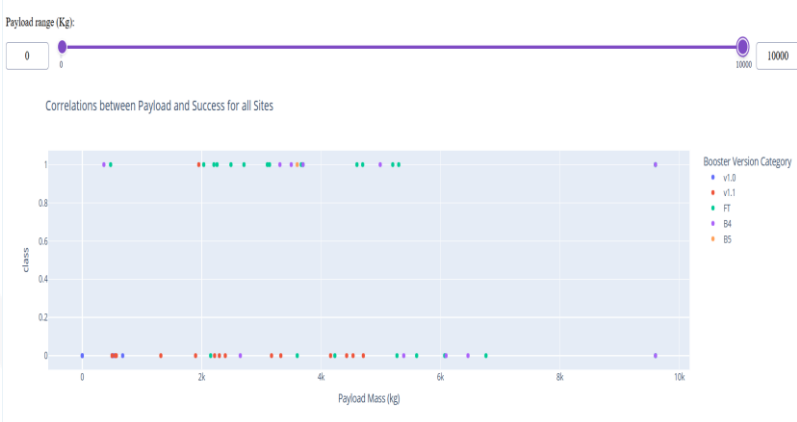
Payload-range filters and booster-version outcomes from the Dash dashboard



0–6,000 kg range



5,000–9,000 kg range



Full payload distribution

Interpretation

The 0–6,000 kg payload range shows stronger successful-launch representation than the 5,000–9,000 kg range. Across the displayed booster categories, FT is identified as the highest-performing version in terms of launch success rate.

Exploratory Data Analysis and Visual Analytics

Investigating launch patterns, site performance, payload characteristics, and booster versions

Data Exploration

- Examined launch records and relevant variables
- Reviewed launch sites, payload mass, orbit type, flight number, and booster version
- Identified patterns and differences across launch conditions

Visual Analysis

- Used charts and interactive dashboard filters
- Compared successful launches across sites
- Examined payload and orbit relationships with landing outcomes

Performance Comparisons

- Calculated success counts and success rates
- Compared booster-version categories
- Investigated how launch characteristics relate to mission success

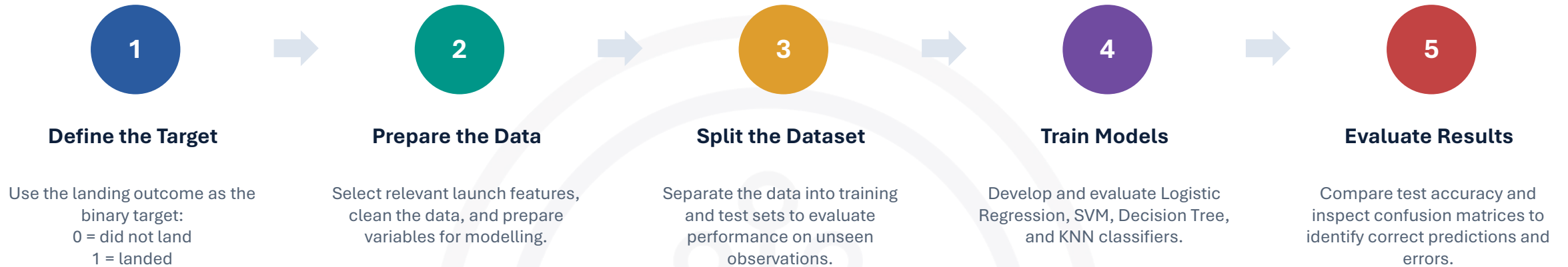
Key Findings from the Visual Analysis

- Launch site** KSC LC-39A recorded the largest share of successful launches and the highest observed site success rate: 76.9%.
- Booster version** The Full Thrust (FT) category showed the highest launch success rate among the evaluated booster versions.
- Payload analysis** Interactive payload-range exploration was used to compare landing outcomes across different payload masses.
- Analytical value** Visual analytics helped reveal differences that would be difficult to identify from raw records alone.



Prediction Analysis Methodology

Machine-learning workflow used to predict Falcon 9 landing outcomes



Models Evaluated

- Logistic Regression
- Support Vector Machine (SVM)
- Decision Tree Classifier
- K-Nearest Neighbours (KNN)

Evaluation Measures

- Test-set accuracy
- Confusion matrices
- Correct versus incorrect classifications
- Comparative assessment of model performance

Objective: identify models capable of reliably distinguishing successful from unsuccessful Falcon 9 landings.



Prediction Analysis: Model Comparison

Evaluation of models predicting Falcon 9 landing outcomes

Logistic Regression

83.3%

15 of 18 correct

Support Vector Machine

83.3%

15 of 18 correct

Decision Tree

83.3%

15 of 18 correct

K-Nearest Neighbours

61.1%

11 of 18 correct

Model	Accuracy	Correct	Observation
Logistic Regression	83.3%	15/18	Strong overall performance
SVM	83.3%	15/18	Matched Logistic Regression
Decision Tree	83.3%	15/18	Matched top accuracy
KNN	61.1%	11/18	Lower predictive performance

Key Interpretation

Logistic Regression, SVM, and Decision Tree achieved identical test accuracies of 83.3%. KNN performed considerably worse at 61.1%. The three higher-performing models correctly classified 15 of 18 test cases.

Note: The SVM confusion matrix was not regenerated because its original training process required several hours.

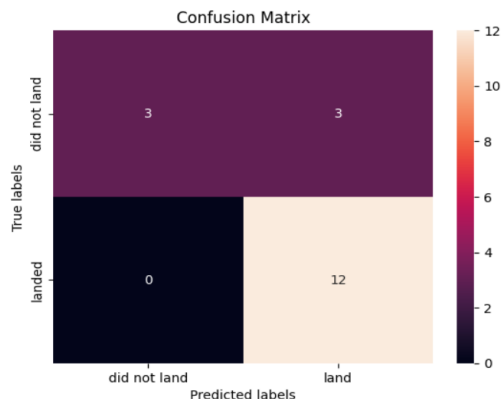
Prediction Analysis: Confusion Matrices

Detailed prediction outcomes for the evaluated classifiers

```
[14]: test_score = logreg_cv.score(X_test, Y_test)
      print(f"(test_score: {test_score})")
      0.833
```

Lets look at the confusion matrix

```
[15]: yhat=logreg_cv.predict(X_test)
      plot_confusion_matrix(Y_test,yhat)
```



Logistic Regression — 83.3%

Logistic Regression

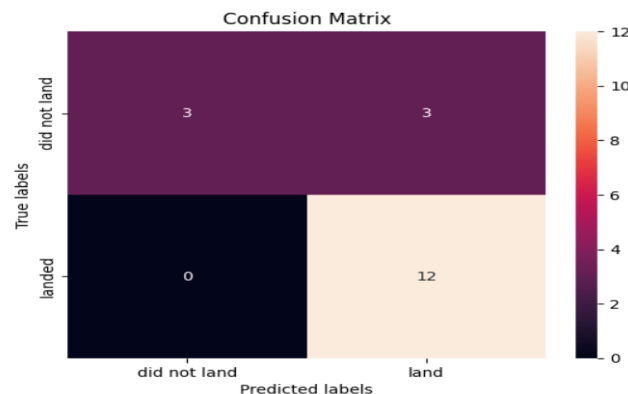
12 landings and 3 non-landings correctly predicted; 3 unsuccessful launches were classified as successful.

Calculate the accuracy of tree_cv on the test data using the method `score` :

```
[21]: test_score = tree_cv.score(X_test, Y_test)
      print(f"(test_score: {test_score})")
      0.833
```

We can plot the confusion matrix

```
[22]: yhat = tree_cv.predict(X_test)
      plot_confusion_matrix(Y_test,yhat)
```



Decision Tree — 83.3%

Decision Tree

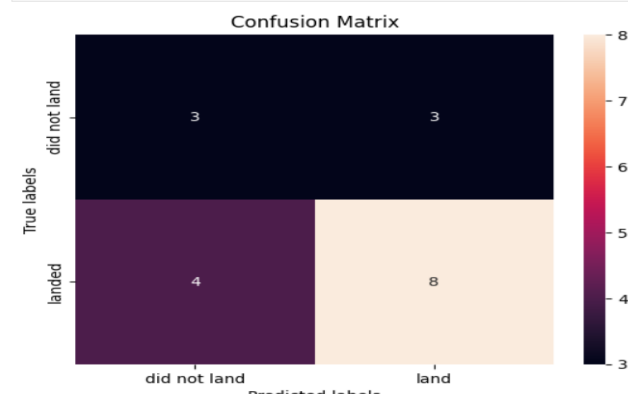
Produced the same confusion-matrix counts and accuracy as Logistic Regression.

Calculate the accuracy of knn_cv on the test data using the method `score` :

```
[26]: test_score = knn_cv.score(X_test, Y_test)
      print(f"(test_score: {test_score})")
      0.611
```

We can plot the confusion matrix

```
[27]: yhat = knn_cv.predict(X_test)
      plot_confusion_matrix(Y_test,yhat)
```



KNN — 61.1%

KNN

Correctly predicted 8 landings and 3 non-landings; 4 successful launches were classified as unsuccessful.



Conclusion and Key Findings

Summary of the exploratory analysis, dashboard findings, and predictive modelling

The analysis shows that launch success varied by launch site, booster version, payload characteristics, and model choice.

Exploratory Analysis

- KSC LC-39A recorded the largest share of successful launches.
- KSC LC-39A also showed the highest launch success rate at 76.9%.
- Payload ranges and booster categories provided additional insight into landing outcomes.

Booster-Version Finding

- The Falcon 9 Full Thrust (FT) booster category recorded the highest launch success rate among the evaluated versions.
- This highlights the importance of examining technology and vehicle-generation differences.

Predictive Modelling

- Logistic Regression, SVM, and Decision Tree each achieved 83.3% test accuracy.
- KNN achieved 61.1% and performed less effectively on this test set.
- The strongest models correctly classified 15 of 18 test observations.

Overall takeaway: Site selection, vehicle configuration, payload characteristics, and model selection all contribute to understanding launch success.



Innovative Insights

Additional observations generated by combining visual analytics with predictive modelling

Launch success is not determined by a single factor; it reflects the interaction of site, payload, vehicle configuration, and operational history.

1

Site Selection as a Strategic Variable

The variation in success rates across launch sites suggests that launch location can be treated as an important feature when analysing mission outcomes.

2

Vehicle Evolution Matters

The strong performance associated with the FT booster category suggests that technological generation may provide useful predictive information.

3

Model Agreement Builds Confidence

Logistic Regression, SVM, and Decision Tree all achieved 83.3% accuracy, indicating consistent performance across different modelling approaches.

4

Accuracy Alone Is Not Enough

The confusion matrices show why model evaluation should include error patterns—not only overall accuracy—especially when missed or false success predictions have different implications.

Practical implication: combine descriptive analytics with predictive models to support more informed launch-planning decisions.

